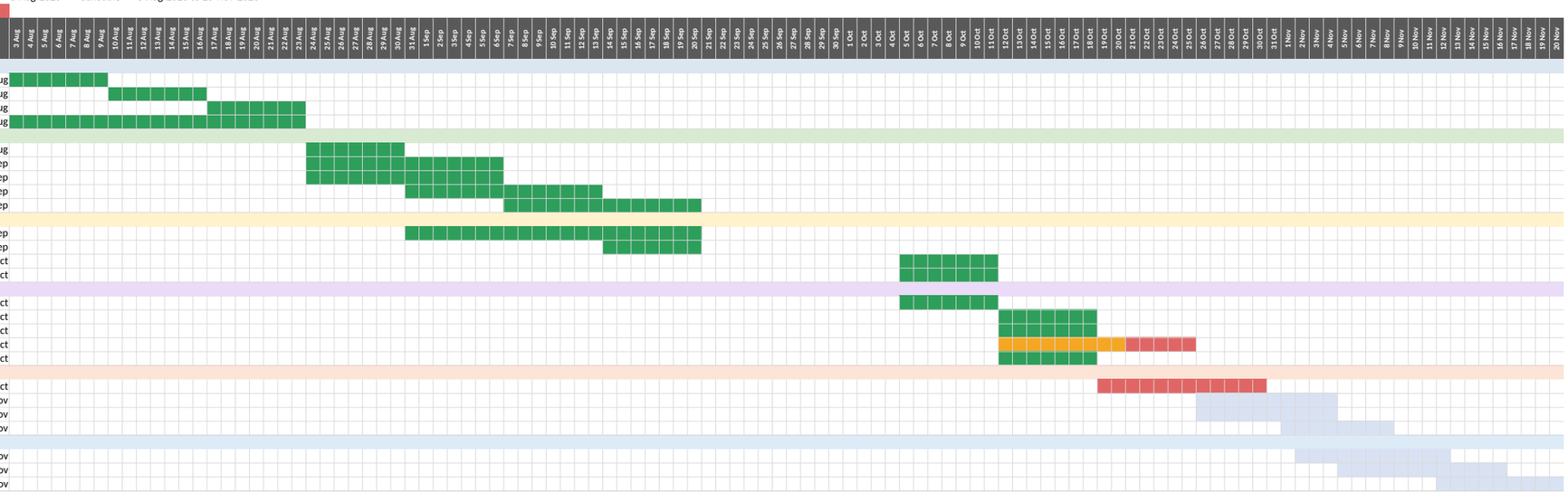
PRISM: Poverty Risk Identification through Statistical Mapping

Explainable Machine Learning Framework for District-Level Poverty and Development Risk Assessment in India

Team: Amish Kapoor | Aryan Chaturvedi | Kushal — Today: 3-Aug-2026 — Schedule: 3-Aug-2026 to 20-Nov-2026

Legend: Completed (100%) | In progress (done part) | Remaining | Overdue

TASK	ASSIGNED TO	PROGRESS	START	END
Phase 1: Problem Definition, Proposal & Initial Design				
Problem Framing, Pain-Point Research & Project Ideation	Kushal	100%	3-Aug	9-Aug
ML Approach Research, Primary Metric & Fusion-Layer Design	Aryan	100%	10-Aug	16-Aug
Proposal Drafting, GitHub Repository Setup & Submission	Kushal	100%	17-Aug	23-Aug
Data-Source/ADP Research, Workflow Design & Pitch Deck	Amish / Aryan	100%	3-Aug	23-Aug
Phase 2: Dataset Collection, Matching & Integration				
Census 2011 Dataset Collection & Name-Variant Review	Kushal	100%	24-Aug	30-Aug
NFHS-5 & MPI 2023 District-Level Data Collection	Amish	100%	24-Aug	6-Sep
ADP Ground-Truth List Collection & District Matching	Aryan	100%	24-Aug	6-Sep
District-Name Normalization, State Key (dkey) & Dataset Merging	Amish / Aryan	100%	31-Aug	13-Sep
Merge Validation, ADP Comparison & Data-Schema Documentation	Kushal / Amish	100%	7-Sep	20-Sep
Phase 3: Data Quality, Feature Engineering & Preparation				
Merged-Dataset EDA: Missing Values, Duplicates & Median Imputati	Amish	100%	31-Aug	20-Sep
Feature Engineering: Literacy, Gender, SC/ST & Worker Rates	Kushal / Aryan	100%	14-Sep	20-Sep
IQR Outlier Detection, Summary & Interpretation	Amish / Aryan	100%	5-Oct	11-Oct
Feature Review, Selection Preparation, Standardization & PCA	Kushal / Aryan	100%	5-Oct	11-Oct
Phase 4: Clustering, Classification & Model Development				
Selection of 12 Clustering Features & Feature Scaling	Amish / Kushal	100%	5-Oct	11-Oct
K-Means and Agglomerative Clustering; ARI Comparison	Amish	100%	12-Oct	18-Oct
Cluster Interpretation, ADP Contingency Table & Analysis	Kushal	100%	12-Oct	18-Oct
L1 Logistic Regression, Random Forest Importance & Stratified CV	Kushal / Aryan	60%	12-Oct	25-Oct
Sequence Diagram & Class Diagram Design and Refinement	Aryan	100%	12-Oct	18-Oct
Phase 5: Model Evaluation, Explainability & ADP Validation				
Classifier Evaluation: F1-Score, Precision, Recall & Confusion Matrix	Kushal	0%	19-Oct	30-Oct
Feature Importance & Explainability Coverage	Aryan	0%	26-Oct	4-Nov
Validation Against Aspirational Districts Programme List	Amish	0%	26-Oct	4-Nov
Robustness, Reproducibility & Final Model Comparison	Kushal	0%	1-Nov	8-Nov
Phase 6: Dashboard, Documentation & Final Delivery				
District-Level Visualization & Results Dashboard	Aryan	0%	2-Nov	12-Nov
Final Report, Results, Limitations & Documentation	Amish	0%	5-Nov	16-Nov
Final Presentation, Viva & Project Demonstration	Kushal	0%	12-Nov	20-Nov